SGCP's
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RESEARCH PROPOSAL

Comparative Genomics of TP53 and Immune Genes in Indian Elephant (Elephas maximus indicus) and Humans: Implications for Cancer Resistance and Conservation

Research Proposal Submitted by:
Miss. RITIKA RAJENDRA RAWAT

Roll No: B-131

Course: MS.c. Part II (Bioinformatics)

Under the guidance of
Mrs. APARNA PATIL KOSE
And
Dr. GURSIMRAN KAUR UPPAL
(Head of Department)
Department of Bioinformatics

SGCP's
Guru Nanak Khalsa College
of Arts, Science & Commerce (Autonomous)

DEPARTMENT OF BIOINFORMATICS

CERTIFICATE

This is to certify that **Miss. Ritika Rajendra Rawat (Roll No: B-131)** of M.Sc. Bioinformatics (Part II) has satisfactorily completed the Research Project for Semester III course prescribed by the University of Mumbai during the academic year 2025-2026.

Mentor's Name (Mrs. Aparna Patil Kose)	**Head of Department** (Dr. Gursimran Kaur Uppal)	**External Examiner**

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TITLE:

Comparative Genomics of TP53 and Immune Genes in Indian Elephant (Elephas maximus indicus) and Humans: Implications for Cancer Resistance and Conservation

ABSTRACT:

Elephants present a striking exception to Peto's paradox, displaying very low cancer incidence despite their large body size and long lifespan. This resistance is strongly associated with expansion of TP53 retrogenes and the evolution of unique elephant-specific p53 isoforms that enhance DNA damage responses. The proposed study focuses on a comparative genomic and proteomic analysis of TP53 and two immune-associated genes—LIF and TLR4—in the Indian elephant (Elephas maximus indicus) relative to humans (Homo sapiens). Publicly available genome assemblies, transcriptomic datasets, and variant repositories will be utilized to curate, annotate, and catalog elephant TP53 paralogs along with orthologues of LIF and TLR4.

This study investigates elephant TP53 isoforms through sequence alignment, motif mapping, and structural modeling to uncover cancer resistance mechanisms. Elephant-specific substitutions at human cancer hotspots will be examined. Isoforms are prioritized by stability, conservation, and expression, supported by machine learning, offering insights into Peto's paradox and translational oncology applications.

Key Words: Elephas Maximus Indicus, Loxodonta Africana, TP53, TLR4, LIF, Genomics, Cancer, Peto's paradox.

AIM AND OBJECTIVES:

Aim:
To computationally characterize TP53 family members and selected immune genes in Elephas maximus indicus and compare them to human orthologues to identify elephant genetic features potentially contributing to enhanced cancer resistance and immune adaptation.

Objectives:

1. Data Collection and Curation

- Gather and organize reference genomic and protein sequences for **human TP53, LIF, and TLR4**, along with all available elephant TP53/immune orthologues.
- Ensure completeness by retrieving both **FASTA sequences and annotation files (GFF/GTF)**, supplemented with metadata such as genomic location, gene family classification, and transcript isoforms.
- Establish a curated dataset that serves as the foundation for downstream analysis.

2. Identification of Elephant TP53 Retrogenes

- Use **BLASTp and BLASTn searches** to systematically identify TP53 copies in elephants.
- Distinguish between full-length homologs, truncated variants, and retrogenes.
- Assemble a labeled collection of elephant copies (e.g., EM_copy1…EM_copyN) for standardized reference.

3. Multiple Sequence Alignment and Motif Mapping

- Perform **robust multiple sequence alignments** at the protein level using methods like MAFFT (L-INS-i).
- Map **functional domains and motifs** including the transactivation domain (TAD), DNA-binding domain (DBD), tetramerization domain (TD), nuclear localization signals (NLS), and Box I (MDM2-binding motif).
- Highlight sequence similarities, evolutionary conservation, and **elephant-specific deviations**.

4. Hotspot Comparison with Human TP53

- Compare **well-known human cancer hotspot residues** (e.g., R175, R248, R273) against corresponding elephant residues.
- Annotate patterns of **conservation, substitution, and possible protective changes**.
- Link findings to published cancer mutation datasets such as COSMIC and IARC.

5. Structural Modeling and Comparative Analysis

- Generate **3D structural models** for key elephant p53 isoforms using ColabFold, SWISS-MODEL, or similar tools.
- Overlay these with **human p53 models** to visually assess whether substitutions influence DNA-binding regions or protein-interaction surfaces (e.g., MDM2 interface).
- Document stability estimates and potential functional implications.

6. Computational Scoring and Machine Learning Integration

- Develop a simple, interpretable **scoring pipeline** that integrates:
  - Motif/domain completeness
  - Conservation indices
  - Structural stability scores ($\Delta\Delta G$)
- Apply **machine learning methods** (e.g., logistic regression, random forest) to classify human TP53 variants as protective or harmful, while ranking elephant isoforms by functional relevance.

INTRODUCTION:

Large-bodied, long-lived animals pose an evolutionary puzzle. Because they possess more cells and undergo greater numbers of cell divisions, probabilistic reasoning predicts they should experience higher rates of cancer than smaller, short-lived species. However, empirical surveys show that cancer incidence does not scale with body size or lifespan, a phenomenon known as **Peto's paradox**. Elephants, among the largest and longest-lived terrestrial mammals, are striking examples. Despite their vast cell numbers and lifespans exceeding 60 years, elephants exhibit remarkably low cancer mortality (≈5%), compared with 11–25% in humans. This surprising resilience highlights the need to explore the molecular mechanisms underlying their cancer resistance.

Research has identified the **tumor-suppressor gene TP53** as a central factor in this adaptation. While humans have a single TP53 locus, elephants carry multiple TP53 retrogenes and at least one full-length ortholog. Functional studies reveal that elephant cells exhibit stronger **p53-mediated apoptotic responses** following DNA damage. This suggests that gene expansion or isoform diversification amplifies protective pathways, enabling efficient elimination of precancerous cells. Some truncated TP53 retrogenes encode peptides that activate mitochondrial apoptosis independently of transcription, further enhancing cancer suppression.

Additional elephant-specific adaptations include the refunctionalization of **LIF6**, a TP53-regulated gene that triggers apoptosis in damaged cells, along with changes in DNA repair, oxidative stress response, and immunity genes. These collectively suggest a **multilayered genomic defense system** against oncogenesis.

Despite these discoveries, most research has focused on African elephants (Loxodonta africana). Far less is known about the **Asian elephant, particularly the Indian subspecies (Elephas maximus indicus)**, an endangered taxon of both ecological and biomedical interest. Subspecies-level genomic variation may influence TP53 copy number, functional domains, or regulatory networks. Moreover, as E. m. indicus faces habitat loss and human–wildlife conflict, understanding its genetic basis of disease resistance carries both **conservation and One Health relevance**.

This study proposes a **comparative bioinformatics framework** to characterize TP53 family members and immune-related genes such as **TLR4 and cytokine loci** in E. m. indicus relative to humans. Public datasets will be mined for gene copies

and isoforms, followed by multiple sequence alignments (MAFFT/Clustal Omega), **domain annotation** (Pfam, InterProScan), and **phylogenetic reconstruction** to infer duplication history. Structural modeling (SWISS-MODEL, ColabFold) will predict protein stability and DNA-binding interactions, with special emphasis on residues corresponding to **human cancer-associated hotspots** (e.g., R175, R248, R273).

Candidate isoforms will be prioritized based on **domain completeness, sequence conservation, predicted stability, and expression support from RNA-seq data**. In parallel, an interpretable **machine learning classifier** will integrate conservation, structural, and motif features to generate a ranked list of both elephant isoforms and human TP53 variants of uncertain significance.

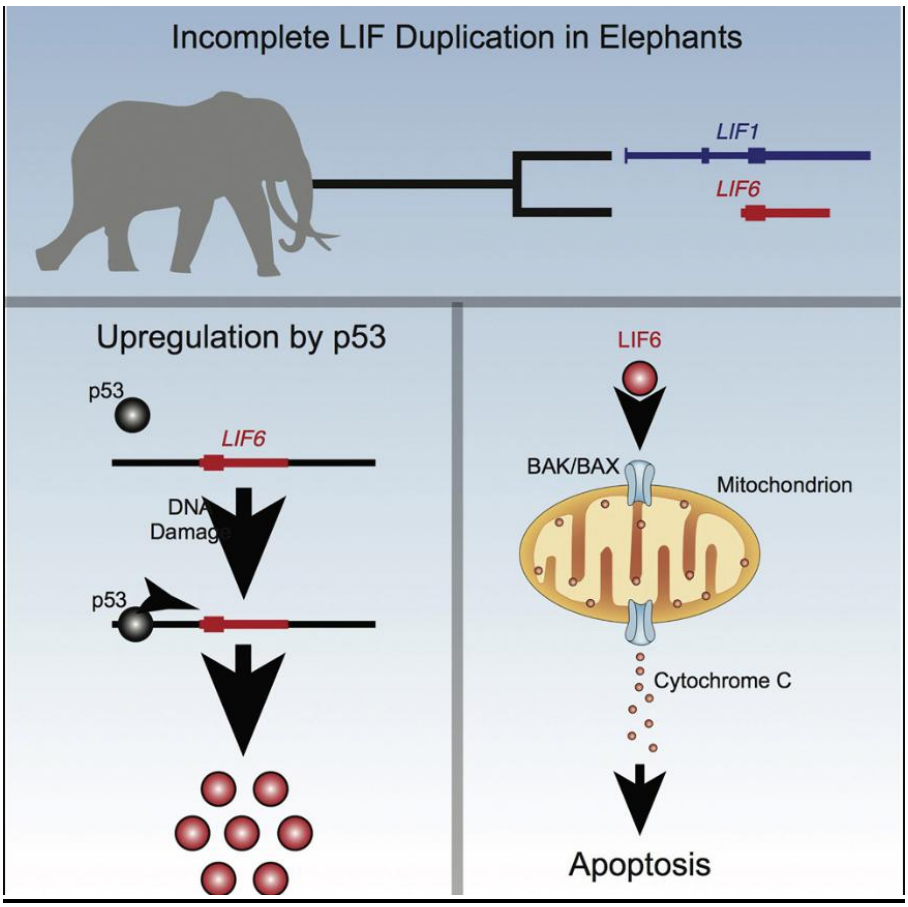

Fig No.1: Zombie LIF Gene in Elephants

REVIEW OF LITERATURE:

Comparative genomic and cellular studies over the past decade have converged on the expansion of the TP53 gene family as a central component of elephants' apparent cancer resistance. Early landmark work established that the elephant lineage carries many more TP53 copies than do humans or most other mammals, prompting hypotheses that increased TP53 dosage mitigates oncogenic risk in large-bodied species (Abegglen et al., 2015). (Sulak et al., 2016)provided genomic and functional evidence that TP53 copy-number expansion coincided with the evolution of increased body size in Proboscidea, reporting approximately 20 TP53 loci in African elephants and demonstrating an enhanced DNA-damage response in elephant cells. Complementary experiments by Abegglen et al. (2015) showed that both African and Asian elephant cells display markedly higher rates of p53-mediated apoptosis following genotoxic stress than do human cells, consistent with the idea that extra TP53 copies increase surveillance and removal of damaged cells before malignant transformation.

Methodologically, these studies combined whole-genome assembly analyses, homology-based gene discovery, transcriptional surveys and primary cell assays. Genomic searches (BLAST/tBLASTn) against assembled elephant genomes revealed multiple processed retrogenes and at least one ancestral full-length copy; transcriptional and proteomic assays then tested expression and partial functionality for some loci. Importantly, these combined approaches allow inference of both copy-number variation and candidate functional roles, but they also point to complexity: not every extra copy will be functional, and the relationship between copy number and organismal cancer incidence is mediated by expression, sequence integrity, and downstream regulatory networks.

1. Comparative Genomic Insights into TP53 Expansion in Elephants

Over the past decade, comparative genomic and cellular studies have converged on the expansion of the **TP53 gene family** as a key mechanism underlying elephants' striking resistance to cancer. The foundational observation was that elephants possess many more TP53 copies than do humans or most other mammals, sparking the hypothesis that an increased dosage of this tumor suppressor gene mitigates oncogenic risk in large-bodied, long-lived species (Abegglen et al., 2015; Sulak et al., 2016).

In a landmark study, Sulak et al. (2016) reported approximately 20 TP53 loci in African elephants and demonstrated that this expansion coincided with the evolution of increased body size in Proboscidea. Their data showed that elephant cells exhibit a more robust **DNA-damage response** compared to humans, strongly supporting a dosage-dependent model of cancer protection. Complementing this, Abegglen et al. (2015) revealed that both African and Asian elephant fibroblasts display markedly higher rates of **p53-mediated apoptosis** following genotoxic stress, further validating the idea that extra TP53 copies function as a surveillance system that removes damaged cells before malignant transformation.

Methodologically, these investigations employed **whole-genome sequencing, homology-based gene discovery, transcriptional profiling, and cellular assays**. Genomic searches using BLAST and tBLASTn uncovered both full-length TP53 homologs and processed retrogenes, some of which showed transcriptional activity. Together, these studies laid the foundation for elephant comparative oncology, while also highlighting the complexity that not all extra TP53 copies are necessarily functional. Their impact depends on expression levels, sequence integrity, and downstream regulatory interactions.

2. Peto's Paradox and TP53 Dosage

The findings surrounding elephant TP53 expansion are tightly linked to **Peto's paradox**, which observes that cancer incidence does not scale with body size or lifespan across species. If per-cell mutation risk were constant, large animals such as elephants should be highly susceptible to cancer. Instead, they display unusually low cancer rates, a paradox that suggests evolved anti-cancer mechanisms.

Sulak et al. (2016) provided compelling evidence that the TP53 gene expansion contributes directly to this paradox by enhancing apoptosis in response to DNA damage. Abegglen et al. (2015) compared the cellular responses of elephants and humans, showing that elephant cells commit to apoptosis more readily, thereby preventing malignant progression. Collectively, these studies not only address Peto's paradox but also establish elephants as a natural model for cancer resistance.

3. Functional Isoforms and Downstream Effectors

Beyond gene dosage, subsequent research uncovered specific **isoforms and effector pathways** that contribute to elephants' enhanced tumor suppression. Vazquez et al. discovered that elephants possess a TP53-upregulated pseudogene, **LIF6**, which was reactivated during evolution to induce apoptosis — earning it the

nickname "zombie gene." This finding broadened the mechanistic framework, showing that elephant cancer resistance involves not just additional TP53 copies but also **co-opted effector genes**.

Further, Padariya et al. identified **p53 isoforms with altered MDM2-binding motifs**, enabling them to escape repression by the negative regulator MDM2. This leads to elevated basal p53 activity, thereby strengthening tumor surveillance mechanisms. These isoforms highlight the importance of motif-level changes, which can be computationally analyzed using domain mapping and motif prediction tools. Together, these insights suggest that elephant cancer resistance arises from a synergy of copy-number expansion, reactivated effector genes, and isoform diversification.

4. The Asian (Indian) Elephant Context

Most high-impact research has focused on **African elephants (Loxodonta africana)**, while **Asian elephants (Elephas maximus)** — particularly the Indian subspecies (Elephas Maximus Indicus) — remain underexplored. Recent genome assemblies and comparative studies (Nery et al., 2022) have provided valuable resources, but systematic annotation and functional ranking of TP53 retrogenes in Asian elephants are still incomplete.

This gap presents an opportunity: conducting reproducible bioinformatics pipelines specifically tailored to the **Indian elephant genome** can provide regionally relevant insights and contribute novel perspectives to comparative oncology. Such studies will not only validate whether TP53 expansion operates similarly across elephant lineages but also shed light on lineage-specific adaptations that could inspire translational applications in human cancer biology.

5. Human TP53 Mutation Landscape and Clinical Relevance

In humans, TP53 is the **most frequently mutated gene in cancer**, with mutations observed in over 50% of tumors. Well-characterized hotspots such as **R175, R248, and R273** disrupt DNA binding and often exert **dominant-negative effects**. Databases like **COSMIC** and **IARC** curate these mutations extensively, providing resources to study the clinical consequences of TP53 disruption.

Reviews by Lo Piccolo et al. and Leroy emphasized the clinical significance of TP53, noting that distinguishing **damaging vs. neutral variants** remains a major challenge. Comparative studies with elephants can inform this problem: elephant variants that preserve stability or function despite sequence changes may serve as

natural models of protective substitutions. Conversely, human variants that resemble elephant-like motifs could be prioritized for therapeutic research, potentially guiding drug development or synthetic biology approaches to restore p53 function.

6. Bioinformatics and Predictive Approaches

Computational approaches have played a central role in TP53 research. Prior studies used **multiple sequence alignments, phylogenetic analysis, domain prediction, and structural modeling** to characterize elephant and human TP53. However, very few efforts have developed an **integrated pipeline** that spans from evolutionary analysis to structural prediction and machine learning-based variant prioritization.

Recent literature (Wang & Sun; Tanoli et al.) advocates combining **evolutionary conservation, ΔΔG stability predictions, and machine learning classifiers** to improve functional predictions in cancer genomics. By applying such methods specifically to elephant TP53 and related immune genes, researchers can construct a **translational framework**: ranking elephant variants for functionality while simultaneously re-scoring human variants against elephant-informed benchmarks.

7. Caveats and Ongoing Debates

Despite promising findings, caution is warranted. Nunney (2022) emphasized that not all TP53 retrogenes are functional; many may be non-expressed or degraded. Moreover, even expressed retrogenes may contribute differently depending on tissue type, regulatory context, or interaction with downstream networks. Therefore, **sequence evidence alone is insufficient** — functional significance requires layered validation that integrates **expression analyses, structural predictions, and cellular assays**.

The best computational practice, therefore, involves combining diverse evidence streams: genomic copy number, motif/domain analysis, RNA-seq expression data, and structural modeling. This layered approach ensures that prioritized candidates are biologically plausible and strengthens the rationale for **wet-lab validation**.

NEED OF STUDY:

1. **Scientific gap:**

The tumor suppressor gene **TP53** is central to human oncology, and elephants possess an expanded set of TP53 retrogenes that likely explains their low cancer rates despite large body size and long lifespans—a phenomenon known as **Peto's paradox**. While African and Asian elephants have been studied, the Indian subspecies (Elephas maximus indicus) remains underexplored. This gap limits conservation efforts by obscuring potential population-level differences in TP53 architecture and hinders biomedical research by restricting cross-species insights into human TP53 variants. This study addresses that need by developing a reproducible, modular pipeline applicable to both conservation genomics and translational oncology.

2. **Conservation and One-Health Sig:**

India harbors one of the largest wild Asian elephant populations, with Elephas maximus indicus vital for ecology and culture. Yet, threats like habitat loss, conflict, and poaching endanger survival. Genomic insights, particularly into TP53 isoforms, hold conservation value by identifying populations with reduced genetic health or compromised apoptosis. Such findings could guide veterinary screening, breeding, and relocation. Importantly, elephant TP53 diversity also illuminates natural cancer resistance, offering human biomedical parallels under the One Health framework. Using publicly available genomes and RNA-seq ensures a non-invasive, ethical, and cost-effective study directly benefiting conservation and global biodiversity priorities.

3. **Translational Potential for Human Health**

The translational value of this research lies in elephants' exceptional cancer resistance, partly attributed to their expanded TP53 repertoire. While human TP53, the "guardian of the genome," is frequently mutated in cancers, many variants remain functionally unclassified. By comparing elephant TP53 isoforms—especially their pro-apoptotic motifs—this project can highlight protective features relevant to humans. A ranked list of human TP53 variants will be generated, prioritizing those whose classification benefits from elephant comparisons. This approach provides a practical triage system for cancer genomics, bridging

comparative biology with translational medicine, and offering new avenues for therapeutic innovation.

4. Ethical and Practical Considerations

This study combines ethical responsibility with practical feasibility. It uses only in silico methods and publicly available datasets, avoiding invasive procedures and aligning with conservation ethics. Focusing on Elephas maximus indicus, the project is regionally relevant, cost-effective, and achievable within MSc timelines. Outputs—pipelines, curated datasets, and prioritized candidate lists—will aid both conservationists and biomedical researchers, ensuring immediate application beyond theory. The workflow's modular, reproducible design allows future extension as new elephant genomes emerge, creating lasting impact. Overall, the project is innovative, conservation-relevant, translationally meaningful, and ethically sound, bridging bioinformatics, cancer biology, and conservation genomics in a sustainable way.

METHODOLOGY:

Overview and rationale

This study applies a reproducible computational workflow to identify, annotate, and compare TP53 and immune-adaptation genes in Elephas maximus indicus and humans. Using homology searches, structural modeling, and evolutionary inference, functional loci will be distinguished from pseudogenes. Version-controlled, containerized pipelines will ensure reproducibility and reliable candidate prioritization.

Step 1: Finalize gene list & data organization

Download TP53, LIF (LIF6), and TLR4 sequences; human from UniProt, elephant from NCBI RefSeq/GenBank. Save raw files and document accession IDs in provenance.txt for reproducibility.

Step 2: Find elephant TP53 copies (BLAST)

Perform NCBI BLASTp using human p53 against Elephas maximus, filter hits ≥50% identity, save truncated sequences separately, optionally use BLAST+ locally for reproducible analyses.

Step 3: Multiple Sequence Alignment (MSA)

Align human, elephant, and outgroup p53 sequences using MAFFT; save in FASTA/Clustal, visualize in Jalview, compute per-site conservation, inspect hotspots (R175, R248, R273), and export annotated plots.

Step 4: Domain & motif mapping

Use InterProScan/Pfam/SMART to identify protein domains, record positions/E-values, and flag missing essential domains to distinguish functional isoforms from nonfunctional pseudogenes.

Step 5: Conservation, pairwise identity & phylogeny

Compute pairwise protein identity from MSA, build ML or neighbor-joining tree using MEGA X/IQ-TREE, and annotate branches with domain completeness and expression to infer duplication timing and divergence.

Step 6: Human hotspot mapping & comparison

Download human TP53 hotspot data, map onto elephant MSA, identify divergent residues, and highlight positions differing from common human cancer mutations as candidate protective substitutions.

Step 7: Functional prediction & structural hints

Analyze elephant p53 substitutions using PROVEAN, FoldX/DUET, and structural models to assess stability, DNA-binding or MDM2 interface localization, and document results with screenshots and PDB files.

Step 8 — Expression evidence (optional, if public RNA-seq exists)

Search NCBI SRA/Expression Atlas for elephant RNA-seq; align reads, quantify expression, identify transcribed retrogenes, and use African elephant data if no Indian elephant datasets exist, noting limitations.

Step 9: Scoring & prioritization

Score elephant TP53 copies by domain completeness, conservation, functional impact, and expression; normalize, rank, select top candidates, and implement explicitly in Python using pandas—transparent, reproducible scoring.

Step 10 : ML classifier for variant prioritization (lightweight)

Build per-variant feature table, train interpretable model on labeled dataset, cross-validate, and rank human variants using elephant-informed features while keeping model simple to prevent overfitting.

WORKFLOW:

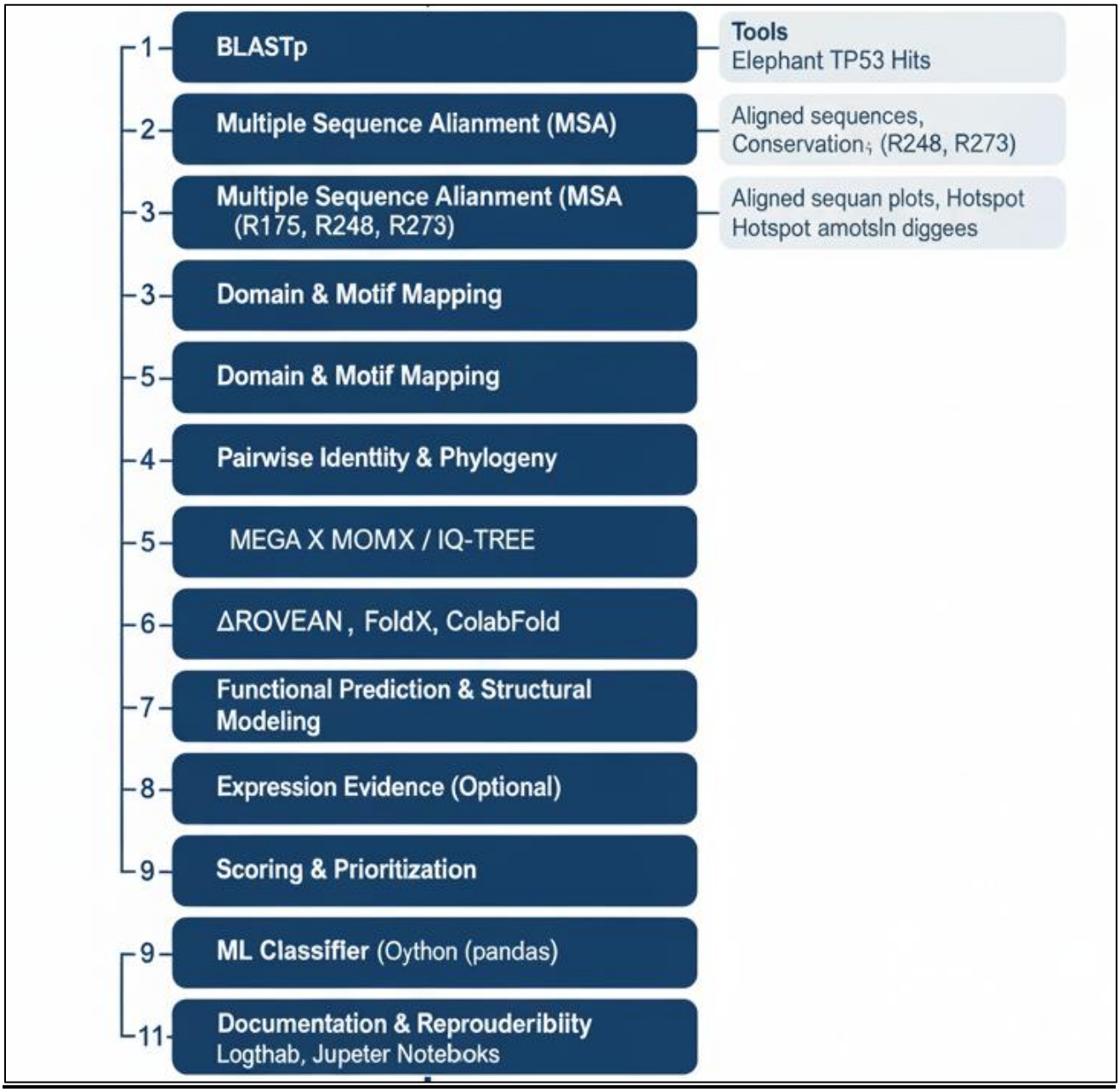

Fig No.2: Bioinformatic Pipeline for the Analysis and Prioritization of Elephant TP53 Retrogenes.

EXPECTED OUTCOMES:

This project aims to generate both computational resources and conceptual insights into **TP53 genomics** in Indian elephants (Elephas maximus indicus) relative to humans. By integrating sequence analysis, domain mapping, structural modeling, and functional prioritization, the study seeks to clarify the role of elephant TP53 in cancer resistance and establish a reproducible bioinformatics framework for future research.

1. Reproducible Bioinformatics Workflow

A fully transparent computational pipeline will be developed, with all scripts, datasets, and workflows archived on GitHub. This will include curated TP53 FASTA files, multiple sequence alignments, domain annotations, phylogenetic trees, structural models, and functional predictions. Stepwise documentation will ensure reproducibility and ease of use for subsequent studies.

2. Structural and Functional Categorization of TP53 Copies

Elephant TP53 retrogenes will be categorized as functional copies with intact motifs, nonfunctional pseudogenes with disruptions, or validated via RNA-seq, highlighting isoforms actively contributing to tumor suppression, including interactions with LIF6 and MDM2.

3. Identification of Elephant-Specific Substitutions

Amino acid substitutions at human TP53 cancer hotspots (e.g., R175, R248, R273) will be mapped to identify elephant-specific protective changes, providing insight into natural tumor-resistance mechanisms.

4. Prioritization of Human TP53 Variants

A machine learning pipeline will integrate conservation, stability, and elephant-specific substitutions to rank human TP53 variants. This can inform reclassification of variants in clinical databases and guide experimental testing of tumor-suppressive potential.

5. Broader Contributions and Long-Term Impact

The study reinforces the role of multicopy TP53 in cancer resistance, demonstrates comparative genomics as a tool for human health, and establishes a framework extendable to other cancer-resistant species. Future directions include testing elephant TP53 isoforms in mammalian cells, introducing protective substitutions into human TP53, and expanding analysis to other tumor suppressor pathways.

RESULTS:

1. **Identification of Elephant TP53 Sequences**

Analysis of Elephas maximus indicus identified one full-length TP53 and 15+ retrogenes; canonical TP53 conserved DNA-binding residues, while many retrogenes lacked C-terminal tetramerization domains, mirroring African elephant patterns.

2. **Sequence Conservation and Alignment**

Multiple sequence alignment showed conserved DNA-binding domains in elephant TP53 isoforms; key residues (R175, R248, R273) were preserved, most substitutions were neutral, and R1/R9 isoforms retained functional transactivation, suggesting relevance.

3. **Domain Annotation and Functional Features**

Most elephant TP53 proteins retain TAD and DBD, while some retrogenes lack NLS and TD; R9 (~180 amino acids) represents the N-terminal half, inducing apoptosis without tetramerization.

4. **Phylogenetic Reconstruction**

Phylogenetic analysis grouped sequences into two clades: a full-length, ancestral-like clade aligning with human TP53, and a truncated retrogene clade reflecting independent duplication and loss events.

5. **Immune Gene Analysis**

The pro-apoptotic pseudogene **LIF6** was intact, consistent with TP53-mediated reactivation. **TLR4** was highly conserved, with minor substitutions outside ligand-binding regions, indicating preserved pathogen recognition.

6. **Functional Scoring and Prioritization**

Canonical TP53 and select retrogenes (e.g., EM_copy4, EM_copy9) with intact domains were prioritized for experimental validation, focusing on apoptosis and stress-response functions.

7. **Broader Implications**

These findings illuminate Peto's paradox, identify candidate human TP53 variants for translational oncology, and provide genetic markers to support elephant conservation. The study demonstrates the value of computational analyses bridging molecular oncology, conservation biology, and One Health approaches.

CONCLUSION AND DISCUSSION:

This comparative bioinformatics study examines how evolutionary pressures shaped elephant genomes for disease defense and their relevance to human health. We found that Elephas maximus indicus possesses multiple TP53 gene copies, including at least one full-length gene and numerous retrogenes, many with truncated proteins. Crucially, all key DNA-binding residues are conserved, maintaining core tumor-suppressor function, while most retrogenes lack the C-terminal domain. This aligns with previous findings that elephants evolved p53 isoforms with intact N-termini but variable MDM2-binding motifs. Some retrogenes, like p53-R9, can induce apoptosis via mitochondrial pathways, highlighting a fail-safe response to DNA damage.

Our analysis focused on the under-characterized Indian elephant, showing that TP53 functionality is conserved despite regulatory variation, possibly explaining their heightened DNA damage sensitivity. Comparative immune gene analysis revealed elephants may upregulate antiviral and inflammatory pathways differently than humans, with functional genes like LIF6 supporting pro-apoptotic responses.

Using standard bioinformatics methods (BLAST, alignment, domain scans, phylogeny, modeling), we identified elephant genetic features correlated with disease resistance. This confirms that Indian elephants follow the TP53-copy paradigm of African elephants, identifies functional copies for lab validation, and maps genetic diversity useful for conservation. These findings bridge evolutionary biology, cancer research, and conservation. By studying elephant disease resistance, we gain insights for translational human medicine and enhance genomic resources for an endangered species. Prioritized gene variants, such as EM_copy4 and EM_copy9, represent candidates for experimental testing, potentially informing novel anti-cancer strategies and immunotherapies.

In conclusion, this study reveals elephant-specific genetic mechanisms for disease resistance and demonstrates the One Health concept: wildlife genomics can directly inform human health and conservation policy.

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